# Akoizo Original Practice - Form 1

### Division C | Circuit Lab | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

**Name: _____ Date: _____**

**1. (2 points) A negative charge's force is…**

Unit: Electrostatics, historical ideas, and event reasoning.
Worked-case data: Two point charges keep their values while separation triples. Predict force.

A. Opposite the electric field

B. Always with the field

C. Always zero

D. Independent of its charge

Answer: _____

**2. (2 points) An ampere is…**

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

A. Joule per coulomb

B. Ohm per meter

C. Coulomb per second

D. Joule per second

Answer: _____

**3. (2 points) Parallel resistors share…**

Unit: Diagrams, nodes, series, parallel, and Ohm's law.
Worked-case data: Find Req for 100 Ω and 300 Ω in parallel on 6 V.

A. Voltage

B. Always current

C. Always power

D. Always resistance

Answer: _____

**4. (2 points) An LED’s forward voltage is…**

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

A. Always zero

B. Equal to any source

C. Device- and condition-dependent

D. Always 0.7 V

Answer: _____

**5. (2 points) XOR(1,1) equals…**

A. 0

B. 1

C. 2

D. Undefined

Answer: _____

**6. (2 points) KCL expresses…**

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

A. Energy creation

B. Resistance addition only

C. Charge conservation

D. Equal current in all branches

Answer: _____

**7. (2 points) A voltmeter connects…**

Unit: Breadboards, multimeters, and a first virtual build.
Worked-case data: A known 100 Ω resistor has 2 V across it; an unknown in series has 4 V. Find the unknown.

A. Across two nodes

B. As an intentional source short

C. Only instead of every wire

D. Only to ground alone

Answer: _____

**8. (2 points) Equal LED current guarantees equal brightness?**

Unit: LED matching, charge-time estimates, and troubleshooting.
Worked-case data: Estimate ideal time for 1.5 Ah supplied at constant 0.30 A.

A. Only for any red LED

B. Only at 5 V

C. No

D. Always

Answer: _____

**9. (2 points) A D flip-flop mainly adds…**

Unit: Transistors, op-amps, storage, and timing.
Worked-case data: An ideal noninverting amplifier has Rf = 9 kΩ and Rg = 1 kΩ. Predict gain.

A. Stored state

B. Unlimited voltage

C. Zero timing constraints

D. Only resistance

Answer: _____

**10. (2 points) Tripling distance changes force to…**

Unit: Electrostatics, historical ideas, and event reasoning.
Worked-case data: Two point charges keep their values while separation triples. Predict force.

A. One third

B. Nine times

C. One ninth

D. Three times

Answer: _____

**11. (2 points) A watt-hour measures…**

Unit: Current, voltage, resistance, power, and energy.
Worked-case data: A 6 V source drives a 300 Ω resistor for 60 s. Find current, power, and energy.

A. Energy

B. Current

C. Resistance

D. Charge alone

Answer: _____

**12. (2 points) Two 100 Ω resistors in series give…**

A. 100 Ω

B. 10000 Ω

C. 200 Ω

D. 50 Ω

Answer: _____

**13. (2 points) A series resistor mainly…**

Unit: Switches, relays, diodes, and LEDs.
Worked-case data: For Vs = 5 V, Vf = 2 V, and R = 300 Ω, estimate current.

A. Limits current

B. Creates charge

C. Guarantees any voltage is safe

D. Reverses polarity

Answer: _____

**14. (2 points) NOT(A AND B) equals…**

Unit: Digital logic and combinational circuits.
Worked-case data: For A=1, B=1, find half-adder sum and carry.

A. (NOT A) AND B

B. A AND B

C. (NOT A) OR (NOT B)

D. A OR B

Answer: _____

**15. (2 points) Negative solved current means…**

Unit: Kirchhoff laws and network problem solving.
Worked-case data: At a node, 12 mA and 8 mA enter. One branch leaves. Find its current.

A. Opposite the assumed direction

B. Impossible circuit necessarily

C. Negative charge disappears

D. No voltage exists

Answer: _____

**16. (2 points) An ohmmeter generally requires…**

Unit: Breadboards, multimeters, and a first virtual build.
Worked-case data: A known 100 Ω resistor has 2 V across it; an unknown in series has 4 V. Find the unknown.

A. Maximum possible voltage

B. No instrument instructions

C. A de-energized suitable circuit

D. A live mains circuit

Answer: _____

**17. (2 points) 1.5 Ah / 0.30 A equals…**

A. 5 h

B. 0.45 h

C. 5 J

D. 0.20 h

Answer: _____

**18. (2 points) The op-amp virtual short requires…**

Unit: Transistors, op-amps, storage, and timing.
Worked-case data: An ideal noninverting amplifier has Rf = 9 kΩ and Rg = 1 kΩ. Predict gain.

A. Saturation

B. Disconnected inputs

C. Suitable negative feedback and linear operation

D. Any circuit whatsoever

Answer: _____